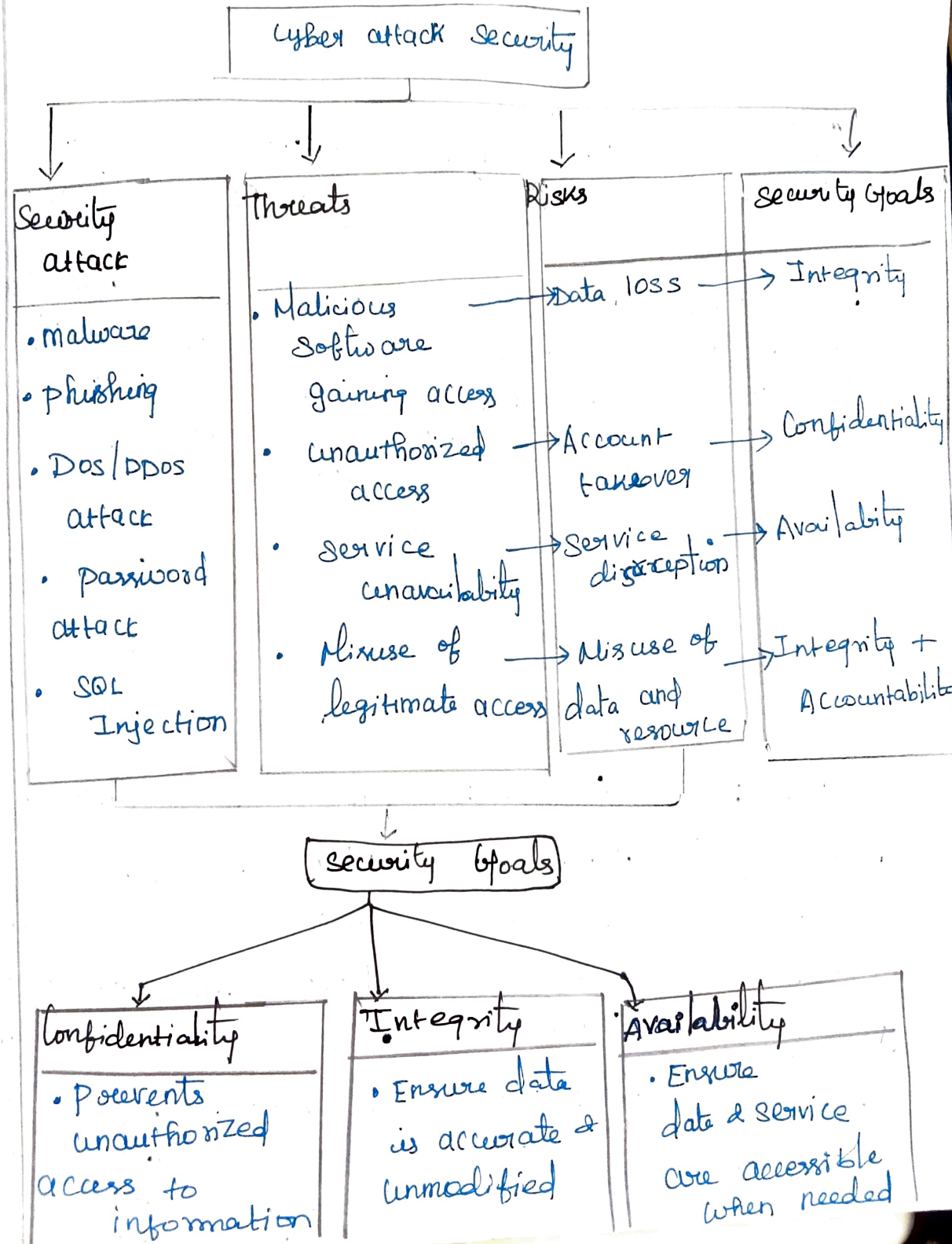
CSA 5111 - CNS

CO1-AT2 - CONCEPT MAP

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1. Create a Concept map linking the Following : Security attacks, Threats, Risks and Security goals



2 Develop a concept map Comparing Substituting Techn... Ceaser cipher, Monoalphabetic cipher, Vigenère Cipher, hill cipher

SUBSTITUTION TECHNIQUES

Caesar Cipher

HOW IT WORKS

- Shifts Each letter by a fixed no. of Position

eg: plaintext: HELLO
shift: 3
ciphertext: KHOOR

Analysis:

- Very simple and easy to break.
- not Secure for real-world use

Monoalphabetic Cipher

HOW IT WORKS

- Each letter is replaced by a different letter (fixed mapping)

eg: plaintext: ATTACK
key: QWERTY...
ciphertext: QFFWE

Analysis:

- More Secure than Caeser
- Still Vulnerable to frequency analysis

Vigenère Cipher

HOW IT WORKS

- uses a key-word to shift letters (different shift for each letter)

eg: Plaintext ATTACK
key: LEMON
ciphertext: LXFOPV

Analysis:

- More secure than Monoalphabetic
- Still breakable using advanced methods

hill cipher

HOW IT WORKS

- Uses Matrix multiplication with a key matrix

Plaintext: AC
key matrix : 3 X 3
ciphertext: POH

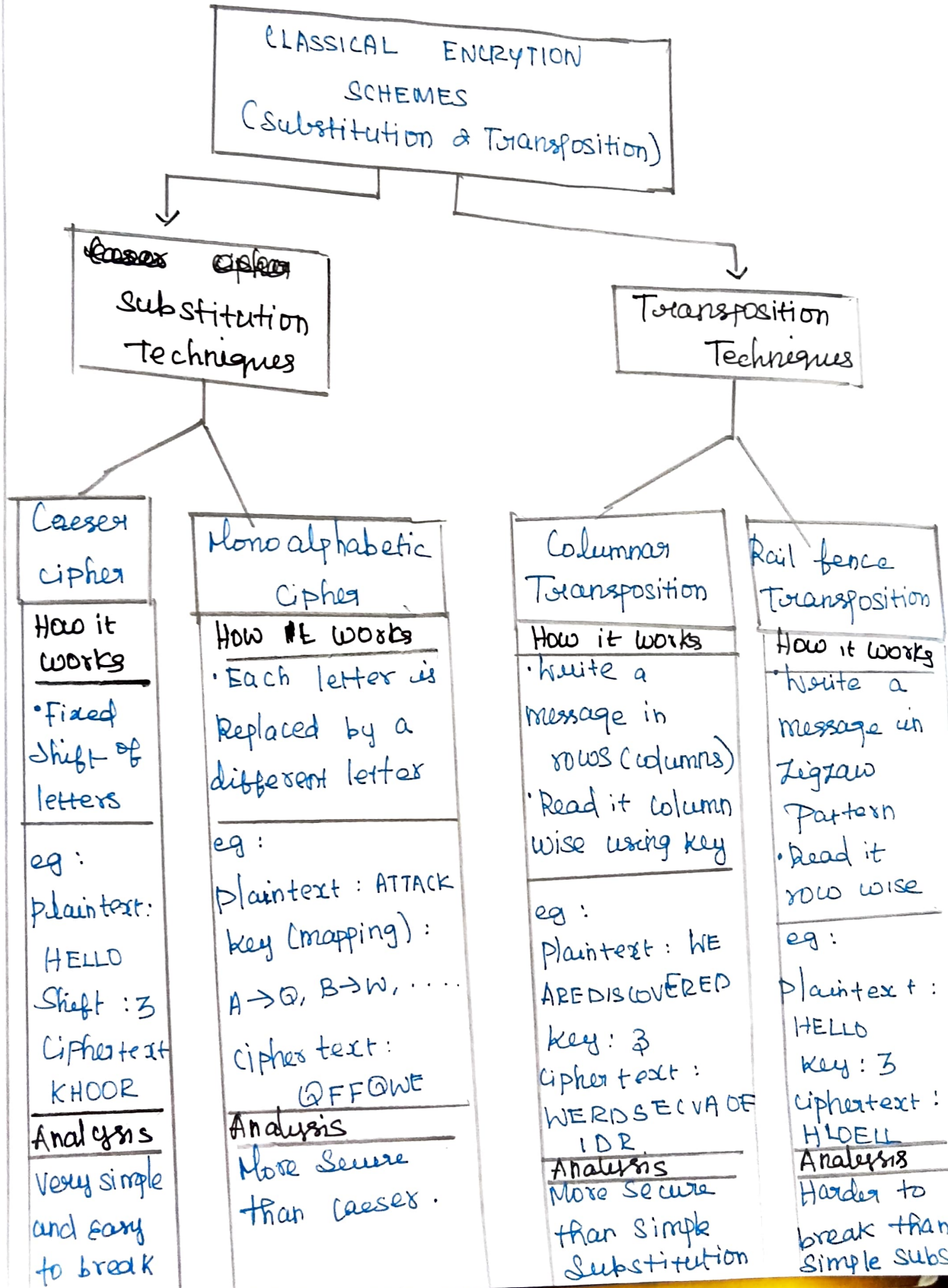
Analysis:

More Secure than caeser and Monoalphabetic

- Vulnerable if the key is Small or weak (ma...

V. Madhumitha

Construct a Concept map for classical encryption Schemes : Substitution Techniques & Transposition Techniques



Create a Concept map Connecting Steganography Concepts Data handling, Watermarking, Survivability

